

Corporate Bond Issuance as a State-Dependent Process: Structural Regimes and Issuance Dynamics in the U.S. (2000–2024)

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Abstract

This paper studies U.S. corporate bond issuance from 2000 to 2024 through a regime-based framework that emphasizes the structural organization of issuance behavior rather than issuance volumes or prices. Using SEC 424B prospectus filings, we construct a monthly panel of issuance structure features capturing participation breadth, issuer concentration, repeated issuance, and temporal compression.

Applying unsupervised learning methods, we identify persistent issuance regimes with distinct structural and dynamic characteristics. A Hidden Markov Model (HMM), estimated exclusively on issuance structure features, reveals four regimes—Normal, Transitional, Window-Grabbing, and Freeze Issuance—characterized by different patterns of participation, coordi-

nation, and timing. Elevated issuance activity is shown to arise from fundamentally different structural configurations, ranging from broadly distributed issuance to highly concentrated and temporally compressed behavior. Window-grabbing episodes are transitory and unstable, while freeze regimes exhibit strong persistence once entered.

Ex post alignment with credit spreads and yield curve conditions shows that similar macro-financial environments can coincide with markedly different issuance behaviors. Overall, the results suggest that corporate bond issuance operates as a state-dependent institutional process shaped by coordination, timing, and access constraints.

1 Introduction

1.1 Why This Matters

Corporate bond issuance plays a central role in the transmission of financial conditions to the real economy. Beyond determining firms' financing costs, the ability to access bond markets shapes investment timing, balance sheet adjustment, and refinancing risk, particularly during periods of macroeconomic stress. Episodes such as the Global Financial Crisis, the COVID-19 shock, and the post-pandemic monetary tightening cycle illustrate that disruptions in issuance activity often precede, accompany, or amplify broader financial instability.

Despite this importance, most empirical analyses of corporate bond issuance focus on aggregate issuance volumes, spreads, or issuance probabilities at the firm level. While informative, these approaches implicitly treat issuance as a scalar outcome—how much is issued or whether issuance occurs—rather than as a collective market behavior that may reorganize structurally under stress. In practice, market participants often describe issuance conditions using qualitative notions such as “windows,” “rush issuance,” or “market freeze,” suggesting that issuance activity may be better understood as operating within distinct structural environments rather than fluctuating smoothly along a single dimension.

Understanding whether corporate bond issuance exhibits systematic regime-like behavior is therefore of first-order importance. If issuance environments differ not only in intensity but also in participation breadth, issuer concentration, repetition, and timing compression, then conventional measures may fail to capture key dimensions of financing risk and market fragility.

1.2 What We Do Not Yet Understand

Existing literature provides rich evidence on how credit spreads, interest rates, and firm characteristics affect issuance decisions. However, three important gaps remain.

First, most studies abstract from the internal structure of issuance activity, treating issuers as independent decision-makers rather than participants in a jointly evolving market environment. This overlooks whether issuance becomes concentrated among a small subset of firms, synchronized across issuers, or temporally compressed into narrow windows.

Second, issuance dynamics are typically analyzed using static or reduced-form frameworks that do not impose discipline on persistence, transition paths, or intermediate states. As a result, it remains unclear whether observed issuance patterns reflect temporary noise or stable structural configurations.

Third, while periods of stress are often identified ex post using macroeconomic or financial indicators, there is limited evidence on whether issuance behavior itself contains forward-looking structural signals that precede or accompany market-wide disruptions.

1.3 What This Paper Does

This paper studies U.S. corporate bond issuance from 2000 to 2024 through the lens of a regime-switching issuance state machine. Rather than modeling issuance volume or firm-level issuance probabilities, we focus on how issuance activity reorganizes structurally over time.

We make three main contributions.

First, we construct a novel monthly panel of issuance behavior based on SEC 424B filings, designed to capture the structure of market participation rather than financing amounts. Our key features measure issuance breadth, issuer concentration, repeated issuance, and temporal compression, jointly characterizing how issuers access the market in each period.

Second, we apply a sequence of unsupervised learning methods to show that issuance behavior clusters into a small number of distinct structural archetypes. A PCA–KMeans framework identifies geometric patterns in the feature space, while a Hidden Markov Model (HMM) imposes temporal discipline, allowing us to study regime persistence and transition dynamics. Importantly, the HMM is estimated on the same standardized features, ensuring that regimes reflect issuance structure rather than macro inputs.

Third, we use a Student-t mixture model as a complementary diagnostic tool. By removing all temporal ordering and Markov structure, the mixture model tests whether issuance archetypes emerge purely from the cross-sectional distribution of features. The alignment between mixture components and HMM regimes demonstrates that the identified regimes are not artifacts of temporal smoothing, but reflect genuine structural organization in issuance behavior.

Together, these results suggest that corporate bond issuance operates as a state-dependent system, cycling between normal participation, transitional adjustment, opportunistic window-grabbing, and market withdrawal. By linking these regimes to macro-financial indicators and crisis episodes, the paper provides a structural framework for interpreting issuance dynamics beyond aggregate volumes or spreads.

2 Data and Sample

2.1 Sample Construction and Coverage

The analysis is based on a monthly panel of U.S. corporate bond issuance activity covering the period from January 2000 to December 2024. The starting point of 2000 reflects the modern institutional structure of the U.S. corporate bond market, characterized by widespread shelf registration, refinancing-driven issuance, and increased issuer stratification. Observations prior to 2000 are retained in the data infrastructure for completeness and pipeline stability, but are excluded from the empirical analysis and interpretation presented in this paper.

Issuance activity is proxied using SEC EDGAR filings of Forms 424B2 and 424B5, which correspond to final prospectus supplements associated with registered bond offerings. These filings provide a comprehensive and time-stamped record of issuers accessing the public bond market. While individual filings do not map one-to-one to realized issuance volumes, they offer a consistent measure of market access and issuance participation, which is the focus of this study.

Raw filings are aggregated to the monthly frequency and linked to issuers using Central Index Key (CIK) identifiers. To minimize survivorship bias, the issuer universe is constructed to include all entities that appear in the filing history over the sample period. No interpolation, smoothing, or gap-filling is applied at any stage of data construction. Months with sparse or negligible activity are retained without adjustment, as irregularity and inactivity are treated as informative features rather than data defects.

All datasets are organized using a reproducible directory structure distinguishing raw, intermediate, and processed data layers. The full panel can be regenerated from source scripts without manual intervention.

2.2 Issuance Structure Variables

Rather than modeling issuance through aggregate volumes or binary issuance indicators, this paper focuses on the structural organization of issuance activity within each month. To capture this structure, we construct a set of issuance features designed to summarize how issuers collectively access the bond market over time.

These features characterize multiple dimensions of issuance behavior, including the breadth of market participation, the concentration of activity across issuers, the prevalence of repeated issuance, and the degree of temporal clustering. Together, they describe whether issuance in a given period is broadly distributed or concentrated, episodic or persistent, and dispersed over time or compressed into narrow issuance windows.

All issuance structure variables are constructed at the monthly frequency and standardized prior to model estimation. By design, these measures abstract from issuance amounts and

instead capture patterns of participation and timing. This allows the analysis to isolate structural reorganization in issuance behavior from fluctuations in aggregate financing demand.

2.3 Macro-Financial Alignment Variables

To place issuance regimes in their broader economic context, the issuance panel is augmented with a set of macroeconomic and financial variables measured at the monthly frequency. These variables include interest rate conditions, credit risk measures, monetary policy indicators, market volatility proxies, and selected real activity indicators.

Importantly, these variables are not used in the estimation of issuance regimes. Regime identification relies exclusively on issuance structure features. Macro-financial variables are introduced only ex post to assess economic consistency, interpret regime transitions, and align issuance behavior with known financial conditions and policy environments.

Daily series are aggregated to monthly frequency using uniform, rule-based procedures without interpolation. All transformations are applied consistently across the sample.

2.4 Final Sample

The final processed dataset consists of a balanced monthly panel covering 2000–2024, combining issuance structure features, macro-financial alignment variables, and regime labels generated by unsupervised models. Summary statistics, variable definitions, and data diagnostics are reported in the Appendix.

3. Methodology

The objective of this paper is not to predict issuance outcomes or estimate causal effects, but to identify whether corporate bond issuance behavior organizes into a small number of structurally distinct and persistent regimes. Accordingly, the methodological choices prioritize interpretability, structural discipline, and internal consistency over predictive accuracy.

The empirical framework builds on standard unsupervised learning and regime-switching methods that are well established in the economics and finance literature. Principal component analysis is used to summarize correlated issuance structure features along their dominant axes of variation (Jolliffe, 2002), while K-means clustering provides a static geometric partition of the resulting feature space (MacQueen, 1967). To impose temporal discipline and study persistence and transition dynamics, issuance regimes are modeled as latent states evolving according to a Hidden Markov Model, following the regime-switching framework introduced by Hamilton (1989). As a complementary diagnostic, the analysis employs a finite mixture of Student-t distributions to characterize the cross-sectional distribution of issuance structures in the absence of temporal dependence. Heavy-tailed mixture models have been shown to provide

robust representations of heterogeneous economic data with extreme realizations (McLachlan and Peel, 2000; Peel and McLachlan, 2000). In this study, the mixture model is used solely as a distributional robustness check rather than as an alternative regime identifier.

The analysis proceeds in three steps. First, a low-dimensional representation of issuance structure is constructed to summarize the joint behavior of issuance features. Second, issuance regimes are modeled as latent states evolving over time using a Hidden Markov Model (HMM). Third, a Student-t mixture model is employed as a complementary diagnostic to assess whether similar structural archetypes emerge in the absence of any temporal structure.

3.1 Feature Representation and Dimensional Structure

Let $\mathbf{x}_t \in \mathbb{R}^K$ denote the vector of standardized issuance structure features observed in month t , where the features summarize participation breadth, concentration, repetition, and temporal compression. All features are constructed at the monthly frequency and standardized to zero mean and unit variance prior to analysis.

Because issuance structure variables are correlated by construction, the analysis begins with a principal component transformation. Specifically, we compute

$$\mathbf{z}_t = \mathbf{W}^\top \mathbf{x}_t,$$

where $\mathbf{W}$ contains the leading eigenvectors of the sample covariance matrix of $\mathbf{x}_t$. The principal components $\mathbf{z}_t$ provide a parsimonious representation of issuance structure while preserving the dominant modes of variation.

The purpose of this step is not dimensionality reduction per se, but to stabilize subsequent regime identification by removing redundant variation and aligning the feature space with its primary structural axes.

3.2 Static Structural Clustering

As a first-pass characterization of issuance structure, we apply K-Means clustering to the principal component scores $\mathbf{z}_t$. Let $c_t \in \{1, \dots, K\}$ denote the cluster assignment at time t , obtained by minimizing within-cluster Euclidean distances.

This static clustering provides a geometric partition of the issuance feature space and serves two roles. First, it offers an interpretable baseline classification of issuance environments without imposing any temporal assumptions. Second, it provides a reference structure against which time-dependent regime models can be aligned and validated. In particular, this step establishes whether economically distinct issuance structures exist in the cross-sectional feature space prior to imposing any assumptions about regime persistence or transition dynamics.

Importantly, cluster assignments are treated as descriptive summaries not as final regime definitions. They do not impose persistence, transition discipline, or dynamic structure.

3.3 Hidden Markov Model for Issuance Regimes

To explicitly model the temporal evolution of issuance behavior, we estimate a Hidden Markov Model in which issuance regimes are represented as latent states that evolve according to a first-order Markov process.

Let $s_t \in \{1, \dots, S\}$ denote the latent issuance regime at time t . The model consists of two components:

1. State transition dynamics

$$\Pr(s_t = j \mid s_{t-1} = i) = \pi_{ij},$$

where $\Pi = (\pi_{ij})$ is an $S \times S$ row-stochastic transition matrix.

2. State-conditional emission distributions

$$\mathbf{z}_t \mid s_t = j \sim \mathcal{N}(\mu_j, \Sigma_j),$$

where μ_j and Σ_j capture the mean issuance structure and dispersion associated with regime j .

The HMM is estimated via maximum likelihood using the Expectation–Maximization algorithm. The number of states S is chosen based on interpretability, stability, and alignment with known issuance patterns, rather than formal information criteria alone.

This framework imposes two economically meaningful constraints. First, issuance regimes are persistent by construction, ruling out spurious month-to-month label switching. Second, regime transitions occur probabilistically, allowing for intermediate or transitional states rather than instantaneous jumps between extremes.

Crucially, the HMM is estimated exclusively on issuance structure features. No macroeconomic or financial variables enter either the transition or emission equations. This ensures that regimes reflect endogenous issuance behavior rather than mechanically encoding external conditions.

3.4 Regime Label Alignment and Interpretation

Because regime labels in unsupervised models are not inherently ordered or named, we align HMM states to the static K-Means clusters based on proximity in the principal component space. This alignment facilitates interpretability and ensures consistency between static and dynamic representations of issuance structure.

Regime labels are assigned ex post based on their dominant structural characteristics and dynamic roles within the issuance process. These labels are descriptive rather than causal and do not encode macroeconomic states a priori.

3.5 Student-t Mixture Model as a Diagnostic Tool

To assess whether issuance regimes arise from the intrinsic structure of the feature space rather than from temporal smoothing, we estimate a Student-t mixture model on the same issuance features.

Formally, the unconditional distribution of $\mathbf{z}_t$ is modeled as

$$p(\mathbf{z}_t) = \sum_{k=1}^M \omega_k t_{\nu_k}(\mathbf{z}_t \mid \mu_k, \Sigma_k),$$

where $t_{\nu_k}(\cdot)$ denotes a multivariate Student-t distribution with degrees of freedom ν_k , and ω_k are mixture weights.

Unlike the HMM, this model imposes no temporal dependence. Each observation is treated as independent, and mixture components are identified purely from the cross-sectional distribution of issuance structure features.

The role of the mixture model is not to define regimes or replace the HMM, but to serve as a diagnostic check. If similar structural archetypes emerge under both models, this provides evidence that the identified regimes reflect genuine organization in issuance behavior rather than artifacts of time-series smoothing.

3.6 Scope and Interpretation

Throughout the analysis, unsupervised models are used as structural inference tools, not predictive devices. Regime labels should be interpreted as summaries of issuance organization rather than forecasts or policy rules. All economic interpretation is introduced only after regime identification, through alignment with macro-financial variables and historical episodes.

4. Empirical Results

4.1 State-Conditional Issuance Structure

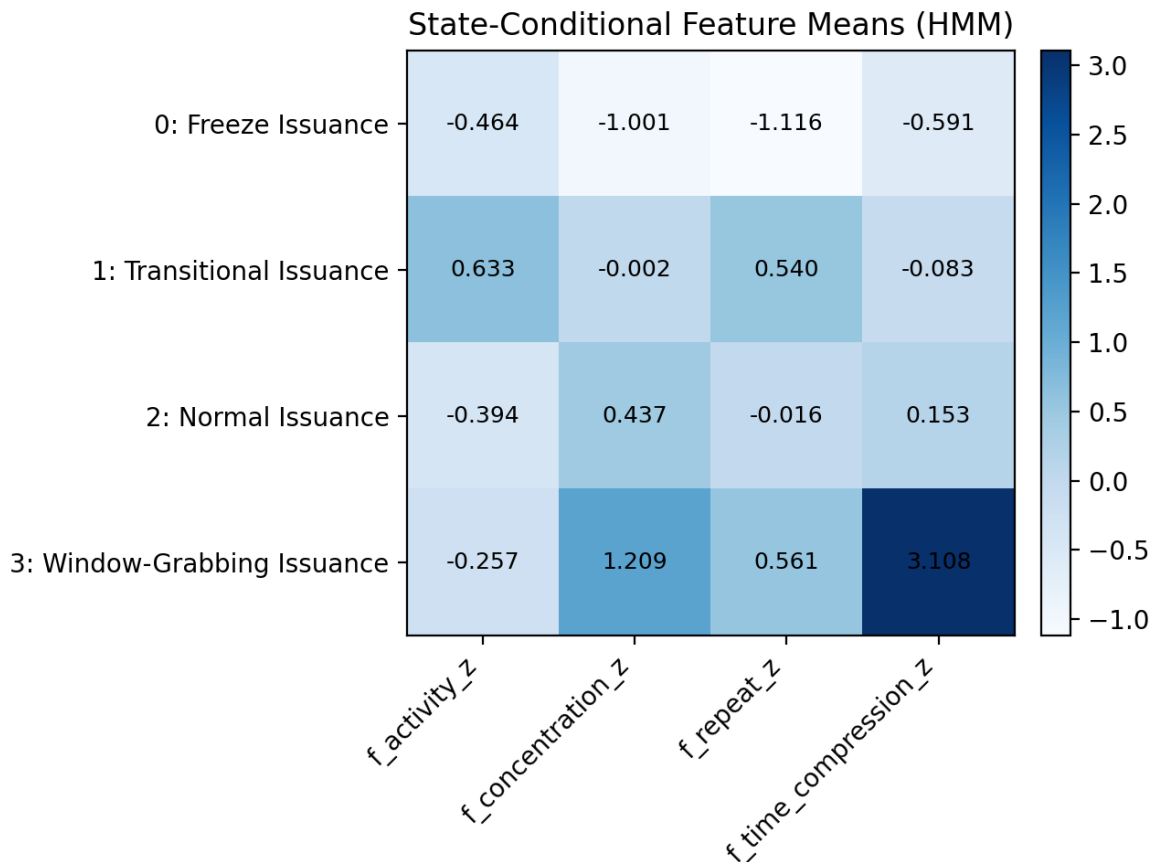


Figure 1: State-conditional issuance structure features.

Figure@fig-hmm-means reports the state-conditional means of standardized issuance structure features estimated by the Hidden Markov Model. The four regimes exhibit sharply differentiated structural signatures across multiple dimensions, confirming that issuance behavior reorganizes along more than a single activity or stress axis.

The Freeze Issuance regime (State 0) is characterized by uniformly negative feature values. Issuance activity is suppressed, repetition is minimal, concentration is low, and temporal compression is weak. This configuration indicates broad withdrawal from issuance activity rather than selective or concentrated access to the market.

The Normal Issuance regime (State 2) occupies a distinct and internally coherent position.

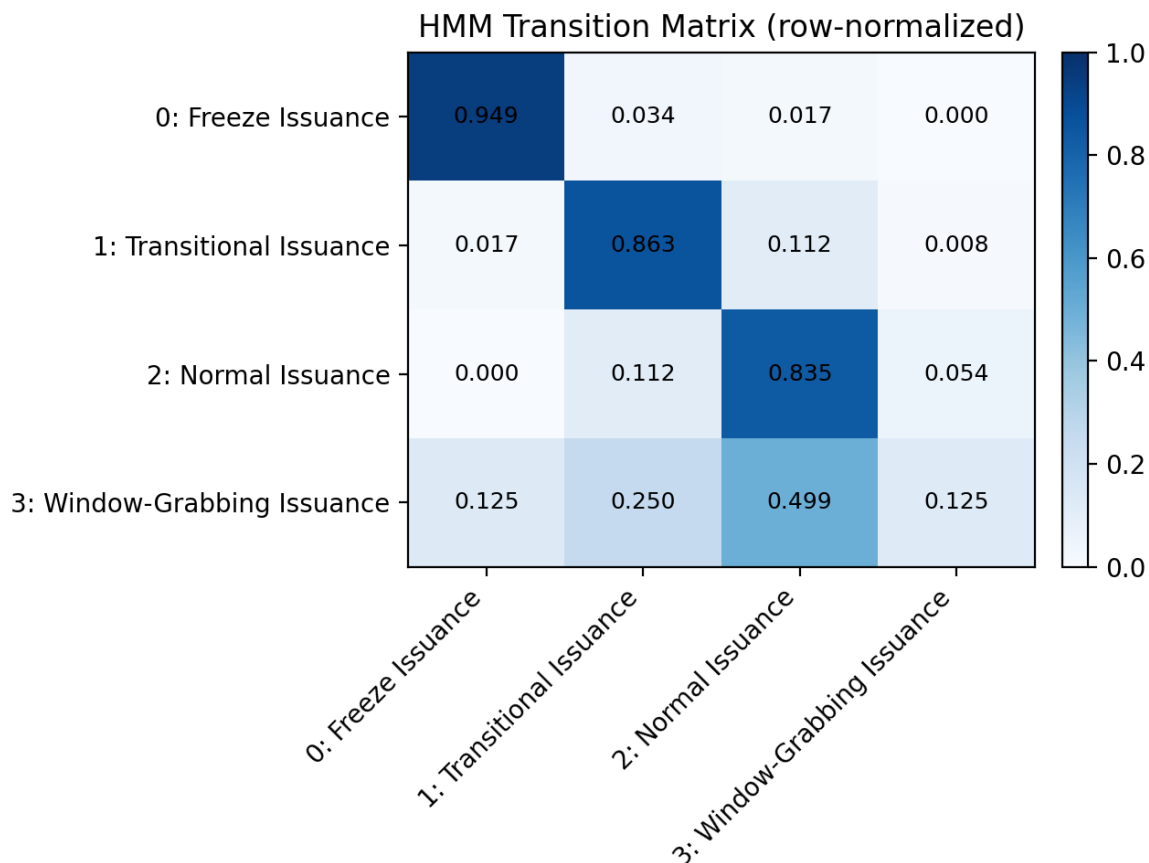
Issuance activity is moderate, concentration remains low, repetition is limited, and issuance is distributed over time. Feature values cluster near zero or slightly positive, consistent with a stable and evenly paced issuance environment. This regime serves as the baseline configuration against which other regimes deviate.

Two regimes display elevated issuance activity but differ sharply in structure. Transitional Issuance (State 1) is marked by higher activity and repetition, while concentration and temporal compression remain muted. This configuration suggests increased issuance intensity without strong coordination or urgency, positioning this regime as an intermediate configuration rather than an extreme state.

By contrast, Window-Grabbing Issuance (State 3) is dominated by extreme temporal compression, accompanied by elevated concentration and repetition. Notably, aggregate activity is not disproportionately high relative to other active regimes. Instead, issuance in this state is distinguished by its timing structure and issuer-level concentration rather than scale alone.

Taken together, the state-conditional feature means highlight that elevated issuance activity can arise from fundamentally different structural configurations. In particular, high activity under dispersed timing and low concentration is empirically distinct from high activity concentrated within narrow issuance windows.

4.2 Regime Persistence and Transition Dynamics



Figure

Figure@fig-hmm-transition reports the row-normalized transition matrix implied by the estimated HMM. The matrix reveals pronounced asymmetries in regime persistence and transition pathways, indicating that issuance behavior evolves along structured and state-dependent trajectories rather than through symmetric or memoryless switching.

The Normal Issuance regime (State 2) exhibits strong persistence, with a high probability of remaining in the same state once entered. This confirms its role as the baseline issuance environment and the central attractor of the issuance system. Similarly, the Freeze Issuance regime (State 0) displays near-absorbing dynamics. Once issuance activity collapses, transitions out of this state are infrequent, reflecting prolonged periods of limited market access rather than short-lived disruptions.

In contrast, the Window-Grabbing Issuance regime (State 3) is markedly unstable. Its self-transition probability is substantially lower than that of other regimes, indicating that episodes of extreme temporal compression and concentration are inherently short-lived. Transitions

out of this state most frequently lead toward the Normal or Transitional regimes rather than persisting or escalating further.

The Transitional Issuance regime (State 1) occupies a distinctive dynamic position. While it exhibits meaningful persistence, it also serves as a frequent conduit between other regimes. Transitions into and out of both extreme states often pass through this intermediate configuration, suggesting that issuance behavior adjusts in stages rather than through abrupt reversals.

Overall, the transition structure implies a staged and path-dependent issuance cycle. Extreme regimes are episodic and unstable, while normal issuance dominates in duration. Adjustments toward and away from extremes are mediated by an intermediate regime that absorbs frictions in issuance timing and coordination.

4.3 Alignment with Macro-Financial Conditions

Having characterized issuance regimes in terms of their structural features and dynamic roles, this section examines how the estimated regimes align with external macro-financial conditions. Importantly, all macro-financial variables are introduced ex post and play no role in regime estimation. The objective is not to identify events or infer causality, but to assess whether the regime classifications exhibit economically coherent patterns when viewed against standard indicators of credit conditions and monetary environment.

4.3.1 Credit Spreads and Market Access

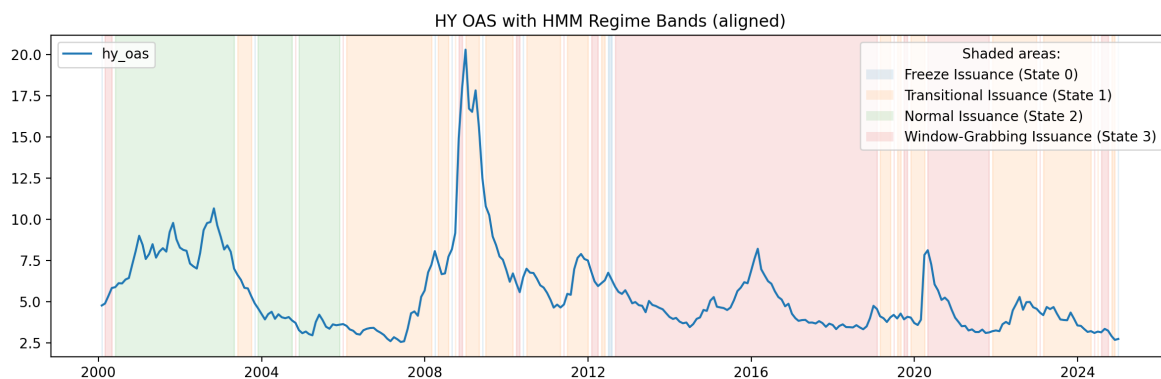


Figure 2: High-yield option-adjusted spreads with HMM regime bands.

Figure@fig-hy-oas overlays the HMM-implied issuance regimes on the time series of high-yield option-adjusted spreads (HY OAS). A clear alignment emerges between issuance structure and the state of credit market access.

Periods classified as Freeze Issuance coincide with elevated and persistent credit spreads. During these intervals, issuance activity is largely absent, consistent with a broad impairment of market access rather than selective or stress-driven issuance. Once this regime is entered, it tends to persist even as spreads fluctuate, highlighting the difference between temporary stress and sustained withdrawal from issuance markets.

In contrast, Window-Grabbing Issuance tends to appear before or during sharp deteriorations in credit conditions, rather than at the peak of credit spreads. Episodes of concentrated and temporally compressed issuance frequently occur while spreads are rising but before they reach extreme levels. This timing pattern indicates that issuance behavior reorganizes in advance of full market shutdown, with firms accelerating access while financing windows remain partially open.

The Normal Issuance regime is predominantly associated with periods of moderate and stable credit spreads. Issuance during these intervals is broadly distributed across issuers and dispersed over time, consistent with a well-functioning primary market. Transitional Issuance occupies the intermediate space, appearing during phases in which credit conditions are adjusting but have not yet resulted in either concentrated issuance or complete market withdrawal.

Overall, the HY OAS overlay suggests that issuance regimes are linked to credit conditions in a nonlinear and state-dependent manner. Rising spreads are associated with changes in the structure and timing of issuance, while persistently high spreads coincide with market withdrawal rather than intensified issuance.

4.3.2 Yield Curve Slope and Monetary Environment

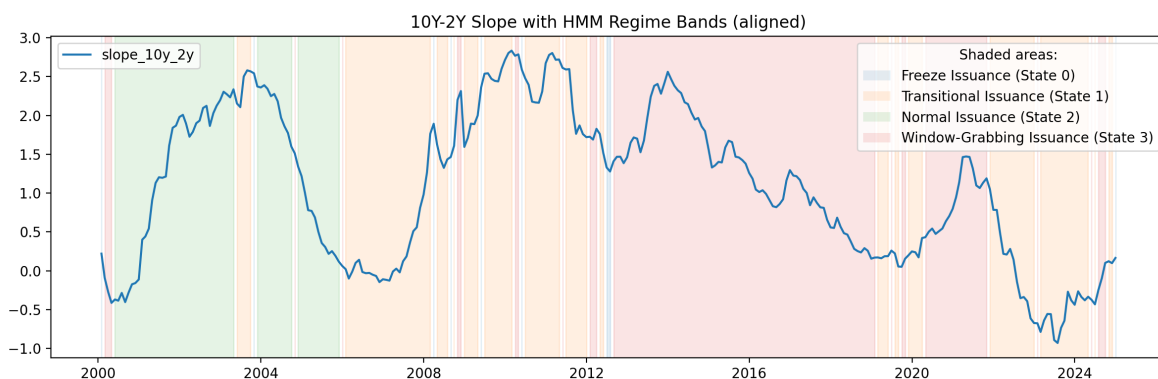


Figure 3: 10Y–2Y Treasury slope with HMM regime bands.

Figure@fig-slope overlays issuance regimes on the slope of the Treasury yield curve, measured as the difference between 10-year and 2-year yields. The yield curve provides a summary indicator of monetary conditions and expectations about future economic activity.

The Normal Issuance regime is most prevalent during periods when the yield curve is positively sloped or moderately flat. These environments are associated with relatively stable financing conditions, under which issuers are able to smooth issuance over time without urgency or coordination pressure.

As the yield curve flattens or approaches inversion, the prevalence of Normal Issuance declines. During these periods, issuance behavior shifts toward Transitional and Window-Grabbing regimes. Issuance activity becomes less evenly distributed over time and more concentrated among a subset of issuers, even when aggregate activity does not collapse.

Sustained or deeply inverted yield curves are more frequently associated with the absence of Normal Issuance and with repeated transitions between Transitional and Window-Grabbing regimes. Unlike credit-spread-driven stress, these environments do not necessarily lead to immediate issuance freezes. Instead, they are characterized by prolonged instability in issuance timing and coordination.

Taken together, the yield curve overlay indicates that monetary environments associated with heightened forward-looking uncertainty disrupt the ability of issuers to maintain a stable issuance pattern, even in the absence of acute credit stress.

4.3.3 Summary

Across both credit spreads and yield curve measures, issuance regimes display consistent and interpretable alignment with macro-financial conditions. Normal issuance dominates in stable environments, window-grabbing emerges as access narrows but remains available, and freeze regimes correspond to sustained impairment of market access. Transitional regimes mediate these shifts, capturing periods of adjustment rather than equilibrium states.

Because macro-financial variables are not used in regime estimation, these alignments arise endogenously from issuance behavior itself, lending credibility to the regime interpretation as reflecting structural organization rather than mechanical conditioning.

5. Diagnostic Validation and Robustness

The issuance regimes identified by the Hidden Markov Model impose explicit temporal structure through persistence and transition dynamics. While this structure is central to interpreting issuance behavior as a state-dependent process, it raises a natural concern: whether the resulting regimes reflect genuine structural organization in issuance behavior, or whether they are artifacts of temporal smoothing or extreme realizations.

To address this concern, we conduct a complementary diagnostic using a Student-t mixture model estimated on the same standardized issuance structure features, but without any temporal ordering or Markov dependence. This approach provides a purely cross-sectional view of

issuance behavior, allowing us to assess whether the HMM regimes align with stable structural archetypes in feature space.

5.1 Alignment Between HMM Regimes and Mixture Components

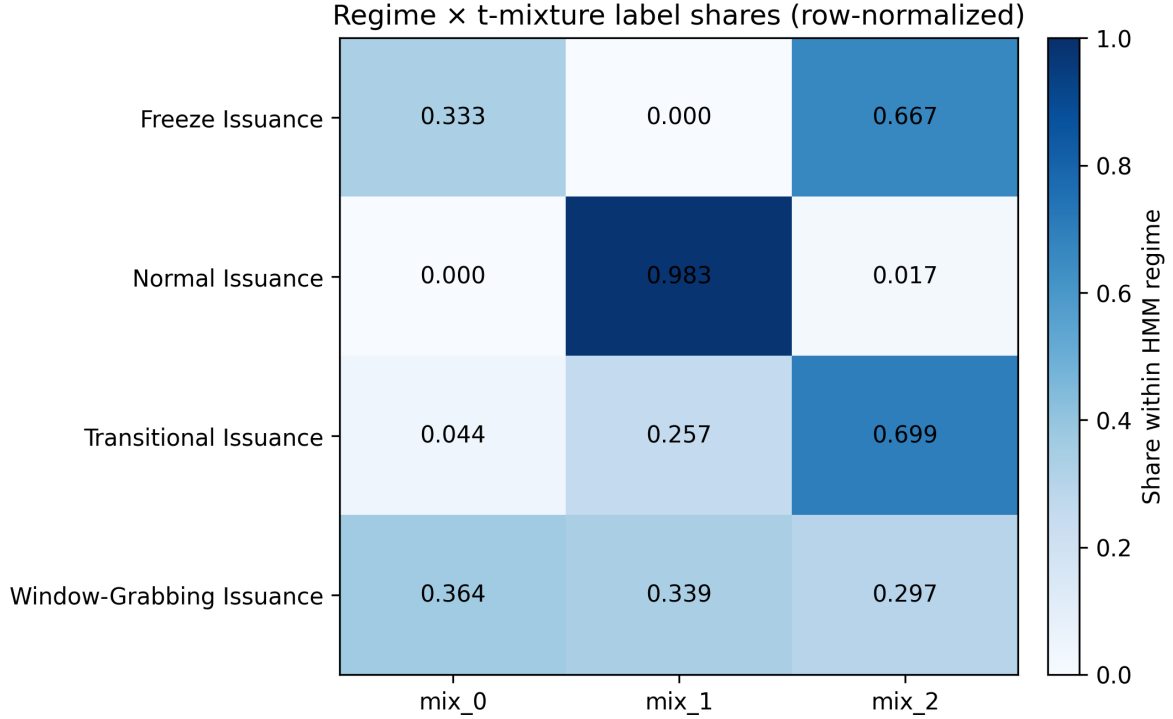


Figure 4: Regime $\times$ t-mixture label shares (row-normalized).

Figure@fig-tmix-heatmap reports the row-normalized distribution of mixture component labels within each HMM regime. A clear and non-degenerate alignment emerges.

The Normal Issuance regime is almost entirely associated with a single mixture component, indicating a high degree of structural coherence. This result suggests that periods of normal market functioning correspond to a well-defined and internally consistent issuance configuration that is stable even when temporal structure is removed.

The Freeze Issuance regime is primarily associated with a different mixture component, reflecting suppressed issuance activity and limited participation. The sharp separation between the Normal and Freeze regimes in mixture space indicates that these regimes are not adjacent variations of a single structure, but represent fundamentally distinct issuance configurations.

By contrast, the Transitional Issuance regime exhibits a mixed composition across components, consistent with its interpretation as an adjustment phase rather than a stable archetype. Ob-

servations in this regime span multiple regions of the feature space, reflecting ongoing reorganization in issuance behavior.

The Window-Grabbing Issuance regime displays the most heterogeneous composition. Its observations are distributed across all mixture components, indicating that while the regime is sharply defined by timing and concentration in the HMM, it does not correspond to a single static cluster when temporal structure is ignored.

5.2 Within-Regime Structural Composition

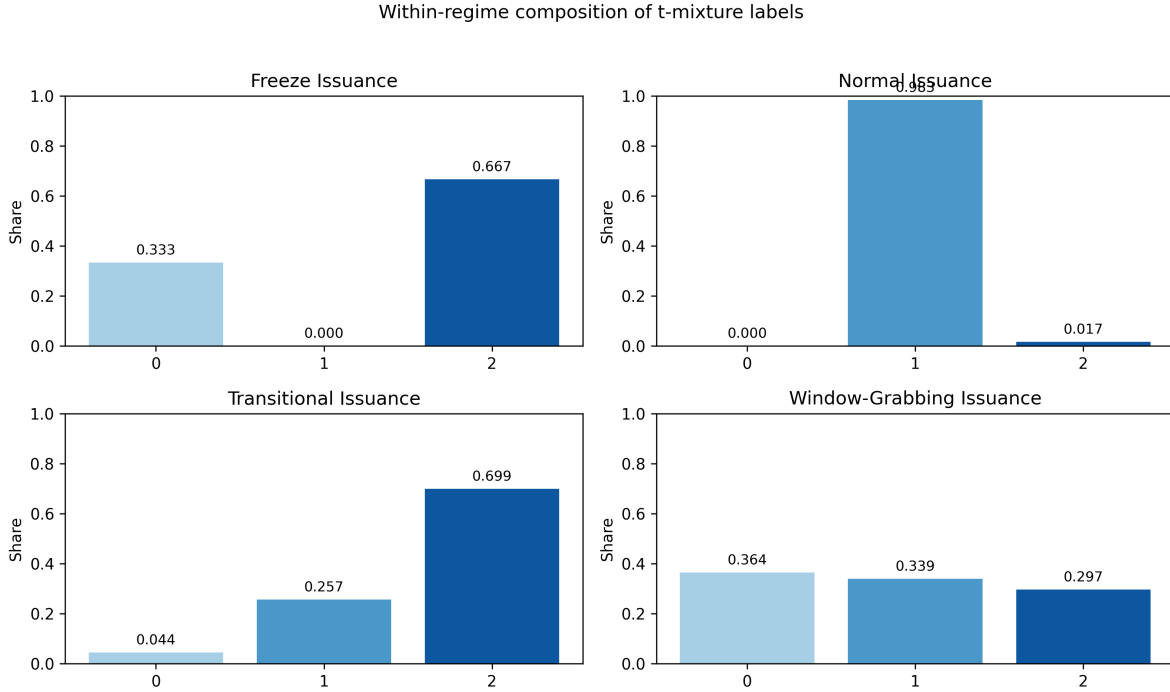


Figure 5: Within-regime composition of t-mixture labels.

Figure@fig-tmix-bars provides a complementary perspective by showing the within-regime composition of mixture components. The contrast across regimes is pronounced.

Normal Issuance exhibits near-complete concentration in a single component, reinforcing its interpretation as the canonical issuance configuration. Freeze Issuance also displays relatively concentrated composition, though distinct from the normal regime, consistent with uniform market withdrawal.

In contrast, Transitional and Window-Grabbing regimes exhibit substantial internal heterogeneity. In particular, the Window-Grabbing regime combines multiple structural archetypes,

suggesting that it represents a dynamic coordination mode rather than a static issuance structure. This heterogeneity is consistent with the transient and episodic nature of window-driven issuance identified in the HMM dynamics.

5.3 Treatment of Extreme Issuance Observations

Throughout the analysis, extreme issuance realizations are retained rather than trimmed or winsorized. This choice reflects the substantive objective of the study, which is to characterize issuance behavior precisely during periods of market stress and episodic, window-driven activity. In such environments, unusually large or concentrated issuance realizations are treated as economically meaningful signals rather than statistical noise.

The modeling framework employed in this study is compatible with this objective. Heavy-tailed mixture distributions accommodate cross-sectional heterogeneity and extreme realizations without imposing arbitrary truncation, while the regime-based dynamics of the HMM allow such observations to be interpreted within persistent and economically coherent issuance states. As a result, extreme observations contribute directly to regime identification and interpretation, rather than distorting them.

6. Discussion

This paper proposes a regime-based perspective on corporate bond issuance, emphasizing how issuance behavior reorganizes structurally under varying degrees of market accessibility. Rather than treating issuance as a smooth or continuous response to financial conditions, the results suggest that firms operate within a small number of persistent behavioral environments that differ in participation, coordination, and timing.

6.1 Issuance Regimes as Behavioral States

A central contribution of the analysis is to formalize issuance regimes as behavioral states rather than as reflections of credit risk or macroeconomic stress per se. Each regime corresponds to a distinct configuration of issuance behavior, characterized by how firms participate in the market, how issuance is distributed across issuers, and how issuance is coordinated over time.

This distinction is important because similar levels of issuance volume or credit spreads can coincide with very different issuance behaviors. The regime framework therefore captures dimensions of market functioning that are not directly observable in price-based indicators, clarifying whether issuance activity reflects decentralized access, selective participation, coordinated acceleration, or broad withdrawal.

6.2 Interpreting Window-Grabbing Issuance

The Window-Grabbing Issuance regime illustrates the value of distinguishing structural behavior from aggregate outcomes. This regime is not defined by unusually high issuance volumes alone, but by a combination of elevated issuer-level repetition, increased concentration, and pronounced temporal compression. Diagnostic evidence further indicates that this regime does not correspond to a single static issuance structure, but instead encompasses heterogeneous configurations united by timing urgency.

These findings suggest that window-grabbing should be interpreted as a coordination response to narrowing market access, rather than as a benign expansion phase. Issuance becomes clustered not because conditions are uniformly favorable, but because firms perceive increasing constraints on future access and adjust issuance timing accordingly. The transitory nature of this regime is consistent with its role as a short-lived adjustment mode rather than a stable operating environment.

6.3 Credit Stress, Policy Uncertainty, and Issuance Adjustment

The regime dynamics highlight an important asymmetry in how issuance behavior responds to different macro-financial environments. Periods associated with acute credit stress tend to coincide with transitions from window-driven issuance toward market withdrawal, reflecting a collapse in effective access. By contrast, environments characterized by prolonged policy uncertainty are associated with frequent reorganization of issuance behavior without immediate shutdown.

This distinction implies that not all forms of financial stress operate through the same behavioral channel. While credit stress constrains access directly, policy uncertainty disrupts firms' ability to smooth issuance intertemporally, leading to unstable or opportunistic behavior even when markets remain formally open. The regime framework provides a structured way to distinguish these environments based on observed issuance behavior rather than on macro indicators alone.

6.4 The Role of Transitional Issuance

The identification of a Transitional Issuance regime further underscores the limitations of binary interpretations of issuance conditions. This regime captures periods in which firms remain active but adjust issuance strategies cautiously, without committing to aggressive timing or withdrawing entirely. Its persistence and frequent appearance along transition paths suggest that a substantial fraction of issuance activity occurs in intermediate states that are neither fully normal nor overtly stressed.

Recognizing this intermediate regime refines the interpretation of issuance cycles by highlighting gradual adjustment and coordination frictions that are obscured in aggregate measures. It

also emphasizes that shifts in issuance behavior often unfold in stages, rather than through abrupt regime changes.

6.5 Regime Dynamics and Structural Interpretation

Finally, the estimated transition dynamics reinforce the interpretation of issuance regimes as components of a structured system. Regimes differ not only in their behavioral characteristics but also in their dynamic roles. Normal issuance dominates in duration, window-grabbing is episodic and unstable, and freeze regimes exhibit strong persistence once entered. Transitional regimes mediate adjustments between these states, absorbing frictions in market access and coordination.

Taken together, the regime-based perspective reframes corporate bond issuance as a dynamic institutional process rather than a sequence of isolated responses to contemporaneous conditions. The framework complements traditional macro-financial indicators by clarifying how issuance behavior is organized, how it adjusts under stress, and how these adjustments evolve over time.

7. Conclusion

This paper studies U.S. corporate bond issuance through a regime-based framework that emphasizes the structural organization of issuance behavior rather than issuance volumes or prices alone. Using issuance structure features derived from SEC filings and a combination of unsupervised learning methods, the analysis identifies a small number of persistent issuance regimes that differ in participation, coordination, and timing.

The results show that issuance behavior does not adjust smoothly to changing financial conditions. Instead, firms operate within distinct behavioral environments that govern how market access is exercised. Normal issuance is characterized by broad participation and dispersed timing, while periods of stress are associated with selective participation, temporal compression, and, in extreme cases, widespread withdrawal from the market. Transitional regimes mediate these shifts, highlighting the importance of intermediate adjustment states that are not well captured by binary notions of “normal” versus “crisis.”

A central finding is the identification of a window-grabbing issuance regime, in which issuance becomes highly concentrated and temporally compressed. This regime is not defined by unusually high issuance volumes alone, nor does it represent a stable equilibrium. Rather, it reflects a short-lived coordination response as firms adjust issuance timing under perceived constraints on future market access. The regime’s transitory nature and heterogeneous structure distinguish it from both normal issuance and complete market freezes.

By separating regime identification from macroeconomic conditioning, the framework provides a structural interpretation of issuance behavior that complements traditional credit market

indicators. Similar levels of spreads or interest rates can coincide with markedly different issuance regimes, underscoring the value of behavioral and organizational measures in understanding market functioning.

Overall, the regime-based perspective reframes corporate bond issuance as a dynamic institutional process shaped by coordination, timing, and access constraints. While the framework is not designed for forecasting or policy evaluation, it offers a systematic way to interpret issuance behavior across different market environments and over time.

8. Limitations and Future Research

This study is subject to several limitations that reflect deliberate design choices rather than methodological shortcomings. First, the analysis focuses on issuance behavior as observed in U.S. corporate bond markets over the post-2000 period. While this window captures multiple stress episodes and policy regimes, extending the framework to earlier periods or to other fixed-income markets may provide additional insights into the generality of the identified issuance regimes.

Second, the regime framework is intentionally designed for structural interpretation rather than prediction. The Hidden Markov Model is used to characterize persistent behavioral states and transition dynamics *ex post*, not to generate real-time signals or forecasts. Future research could explore how regime information might be incorporated into predictive or decision-oriented settings, subject to appropriate validation.

Third, the issuance structure features emphasize timing, concentration, and participation, abstracting from security-level pricing, maturity choice, and covenant design. Integrating issuance behavior with contract characteristics could further enrich the interpretation of regime dynamics and help connect behavioral states to financing costs and risk-sharing outcomes.

Finally, while the analysis identifies distinct regimes in issuance behavior, it does not attempt to assign causal interpretations to macroeconomic or policy variables. Future work could investigate how specific institutional changes, regulatory interventions, or monetary policy actions interact with issuance regimes, building on the descriptive structure documented here.

Appendix A. Data Sources and Variable Dictionary

This appendix documents the raw variables, derived factors and macro-financial series used in the analysis. It emphasises transparency and reproducibility, while clearly distinguishing the information used for regime estimation from variables that are introduced only *ex post* for economic alignment. Throughout, all data are aggregated to the monthly frequency and no imputation, smoothing or gap-filling is applied, consistent with the principle that sparse or extreme observations are informative about issuance behaviour.

A.1 Variable categories

A.1 Variable categories

Issuance counts (SEC EDGAR 424B2/B5; monthly). Variables include issuance event counts and issuer counts at the issuer-month level. `issuer_count` feeds into the activity factor; other count variables are retained as descriptive diagnostics.

Issuer concentration (SEC EDGAR; monthly). Concentration is summarized by `avg_filings_per_issuer`, which is used to construct the concentration factor.

Repeat issuance and timing (SEC EDGAR; monthly). Repeat behavior and issuance timing are captured by `issuer_repeat_ratio` and `median_days_since_last_filing`, which enter the repeat and time-compression factors.

Shelf preference (SEC EDGAR; monthly). The shelf-registration share `share_424B5` is included for descriptive context and is not used in regime estimation.

Engineered factors (derived; monthly). The standardized issuance factors `f_activity_z`, `f_concentration_z`, `f_repeat_z`, and `f_time_compression_z` are the exclusive inputs to PCA, K-Means, the HMM, and the Student-t mixture model.

Macro-financial alignment variables (FRED / U.S. Treasury / CBOE; monthly). Treasury yields, yield-curve shape measures, credit yields and spreads, monetary policy indicators, volatility proxies, and real activity series are appended only ex post for economic interpretation and do not enter regime estimation.

The engineered factors are standardized versions of selected issuance variables and are the exclusive inputs to the unsupervised models. Macro-financial variables are appended only after regime identification for economic alignment and interpretation.

A.2 Construction notes

- **Data aggregation.** SEC prospectus filings (Forms 424B2 and 424B5) are linked to issuers via Central Index Key identifiers and aggregated to monthly counts. Unique issuers, filing repetition and gaps since the previous filing are computed at the issuer-month level. For macro series, daily observations are aggregated to monthly averages or month-end values using calendar-consistent rules without interpolation.
- **Window definitions.** Three temporal windows are maintained: a raw ingestion window (1990–2024), an analysis window (2000–2024) used for model estimation, and a pre-2000 window used only for initialization. Observations prior to January 2000 are excluded from interpretation but retained to stabilize lagged calculations.

- **Variable role.** Only the standardized issuance factors ($f_activity_z$, $f_concentration_z$, f_repeat_z , $f_time_compression_z$) enter the PCA, K-Means, Hidden Markov Model and Student-t mixture. All macro-financial variables are kept separate and are introduced solely for ex post economic interpretation; they do not affect clustering, state estimation or transition dynamics.

Appendix B. Issuance Structure Variable Construction

This appendix formalizes the construction of the issuance structure factors used in the analysis. Each factor summarizes a distinct dimension of market access without reference to issuance amounts or prices. After construction, all factors are standardized to z-scores within the analysis window to ensure comparability across variables.

B.1 Activity (breadth)

The activity factor measures the breadth of market participation. For month t , let I_t denote the number of distinct issuers that filed a prospectus. The raw activity measure is defined as

$$f_{activity,t} = I_t.$$

A higher value indicates a broader set of issuers accessing the market. No winsorization or trimming is applied; unusually high issuer counts during rush periods are retained by design.

The standardized activity factor is computed as

$$f_{activity,t}^z = \frac{f_{activity,t} - \bar{f}_{activity}}{\sigma_{f_{activity}}},$$

where the mean and standard deviation are computed over the analysis window.

B.2 Concentration

The concentration factor captures whether issuance is evenly distributed across firms or concentrated among a small set of repeat issuers. Let F_t denote the total number of filings and I_t the number of issuers in month t . The average number of filings per issuer is

$$\bar{n}_t = \frac{F_t}{I_t}.$$

The concentration factor is therefore defined as

$$f_{\text{concentration},t} = \bar{n}_t.$$

High values indicate that one or a few issuers account for a disproportionate share of filings, signaling a more concentrated issuance environment. This raw measure is standardized to obtain $f_{\text{concentration},t}^z$.

B.3 Repeat issuance

The repeat factor quantifies the extent of multiple filings by the same issuer within a month. For each issuer-month pair, let $n_{i,t}$ denote the number of filings, and define an indicator $\mathbf{1}\{n_{i,t} > 1\}$ for repeat issuance. The repeat ratio in month t is given by

$$f_{\text{repeat},t} = \frac{1}{I_t} \sum_{i=1}^{I_t} \mathbf{1}\{n_{i,t} > 1\}.$$

This variable ranges between 0 and 1 and reflects refinancing pressure or opportunistic repeated issuance. After computation, $f_{\text{repeat},t}$ is standardized to obtain $f_{\text{repeat},t}^z$.

B.4 Temporal compression

Issuers sometimes cluster filings into very short windows, resulting in “window-grabbing” episodes. To capture this behavior, the temporal compression factor is constructed from the gap between consecutive filings. Let $\text{gap}_{i,t}$ denote the number of days since issuer i ’s previous filing (measured across the full sample), and let med_gap_t be the median of $\text{gap}_{i,t}$ across issuers active in month t .

B.5 Standardization

For each factor $f \in \{f_{\text{activity}}, f_{\text{concentration}}, f_{\text{repeat}}, f_{\text{time_compression}}\}$, the standardized value is computed as where $\bar{f}$ and σ_f denote the mean and standard deviation over the analysis window. These standardized factors enter the PCA, K-Means clustering, Hidden Markov Model, and Student-t mixture described in Appendix C.

Appendix C. Model Specification and Estimation Details

This appendix presents the mathematical specification of the regime models used in the analysis and describes key estimation choices. Interpretation of the regimes is deferred to the main text; the focus here is solely on model structure and estimation methodology.

C.1 Hidden Markov Model

Let

$$\mathbf{z}_t = (f_{\text{activity},t}^z, f_{\text{concentration},t}^z, f_{\text{repeat},t}^z, f_{\text{time_compression},t}^z)^\top$$

denote the four-dimensional vector of standardized issuance structure factors observed in month t . A Hidden Markov Model (HMM) with S latent states describes the evolution of issuance regimes as follows.

State transitions. The latent state $s_t \in \{0, \dots, S-1\}$ follows a first-order Markov process with transition matrix $\Pi = (\pi_{ij})_{i,j=0}^{S-1}$, where

$$\Pr(s_t = j \mid s_{t-1} = i) = \pi_{ij}, \quad \sum_{j=0}^{S-1} \pi_{ij} = 1.$$

Emission distribution. Conditional on the latent state, the observed vector $\mathbf{z}_t$ follows a multivariate normal distribution,

$$\mathbf{z}_t \mid s_t = j \sim \mathcal{N}(\mu_j, \Sigma_j),$$

where $\mu_j \in \mathbb{R}^4$ and Σ_j is a full 4×4 covariance matrix. This specification allows the issuance structure factors to be correlated within a regime.

Initial distribution. The initial state s_1 follows a categorical distribution with probabilities π_0 .

The HMM parameters $(\pi_0, \Pi, \{\mu_j, \Sigma_j\}_{j=0}^{S-1})$ are estimated by maximum likelihood using the Expectation–Maximization (Baum–Welch) algorithm. Random initializations are used for means, covariances, and transition probabilities. Iteration proceeds until the change in log-likelihood falls below 10^{-6} or a maximum of 500 iterations is reached.

In this study, $S = 4$ states are selected based on interpretability and dynamic stability rather than formal information criteria. After estimation, HMM state labels are aligned to static K-Means clusters using a Hungarian assignment based on the proximity of state-conditional means, ensuring consistent regime ordering across models.

C.2 Student- t Mixture Model

As a diagnostic check, a finite mixture of multivariate Student- t distributions is estimated on the same standardized factors $\mathbf{z}_t$, ignoring any temporal ordering. The mixture contains K components with fixed degrees of freedom ν and diagonal covariance matrices. The unconditional density is

$$p(\mathbf{z}_t) = \sum_{k=1}^K \omega_k t_\nu(\mathbf{z}_t \mid \mu_k, \text{diag}(\sigma_k^2)),$$

where $\omega_k \geq 0$ are mixture weights satisfying $\sum_{k=1}^K \omega_k = 1$, and $t_\nu(\cdot \mid \mu, \text{diag}(\sigma^2))$ denotes the four-variate Student- t density with mean μ , diagonal scale σ , and degrees of freedom ν . The mixture parameters $(\omega_k, \mu_k, \sigma_k^2)$ are estimated using an Expectation–Maximization algorithm. In the E-step, latent component responsibilities are computed, and in the M-step, mixture weights, means, and variances are updated. The degrees-of-freedom parameter is fixed at $\nu = 6$, and convergence is declared when successive log-likelihood improvements fall below 10^{-6} .

In the analysis, $K = 4$ components are chosen to facilitate comparison with the four HMM regimes. Component labels are post-aligned to HMM states by maximizing the diagonal mass of the contingency table between mixture assignments and HMM regime labels.

Appendix D. Supplementary Regime Results

The main text discusses the qualitative characteristics and economic interpretation of the four issuance regimes identified by the Hidden Markov Model. This appendix reports the corresponding numerical results for completeness.

D.1 State-conditional means

Table D.1 reports the mean of each standardized issuance structure factor within each HMM state. State numbering follows the convention used throughout the paper: 0 = Freeze Issuance, 1 = Transitional Issuance, 2 = Normal Issuance, and 3 = Window-Grabbing Issuance. All entries are z-score means; positive (negative) values indicate above- (below-) average behavior along the corresponding dimension. These values correspond to the state-conditional heatmap shown in Figure@fig-hmm-means.

Table D.1. State-conditional means of issuance structure factors

State	Regime name	Activity (z)	Concentration (z)	Repeat (z)	Time compression (z)
0	Freeze Issuance	-0.464	-1.001	-1.116	-0.591
1	Transitional Issuance	0.633	-0.002	0.540	-0.083
2	Normal Issuance	-0.394	0.437	-0.016	0.153
3	Window-Grabbing Issuance	-0.257	1.209	0.561	3.108

D.2 Transition probabilities

Table D.2 reports the row-normalized transition matrix inferred from the HMM. Each row corresponds to the current state and each column to the subsequent state; entries denote $\Pr(s_{t+1} = j \mid s_t = i)$. Rows sum to one by construction.

Table D.2. HMM transition matrix (row-normalized)

	From	To	0: Freeze	1: Transitional	2: Normal	3: Window-Grabbing
	0: Freeze		0.949	0.034	0.017	0.000
	1: Transitional		0.017	0.863	0.112	0.008
	2: Normal		0.000	0.112	0.835	0.054
	3: Window-Grabbing		0.125	0.250	0.499	0.125

The transition matrix exhibits pronounced asymmetries. Freeze Issuance is nearly absorbing once entered, Normal and Transitional regimes display strong persistence, and Window-Grabbing episodes tend to transition quickly to either Normal or Transitional issuance.

D.3 Regime duration distribution

Figure D.1 illustrates the distribution of contiguous episode lengths (in months) for each regime based on the aligned HMM state sequence. Table D.3 summarizes approximate duration statistics inferred from the figure.

Table D.3. Approximate regime duration statistics

State	Regime name	Mean (months)	Median (months)	Approx. range (months)
0	Freeze Issuance	1	1	1–1
1	Transitional Issuance	5	5	1–15
2	Normal Issuance	13	13	11–36
3	Window-Grabbing Issuance	2	2	1–3

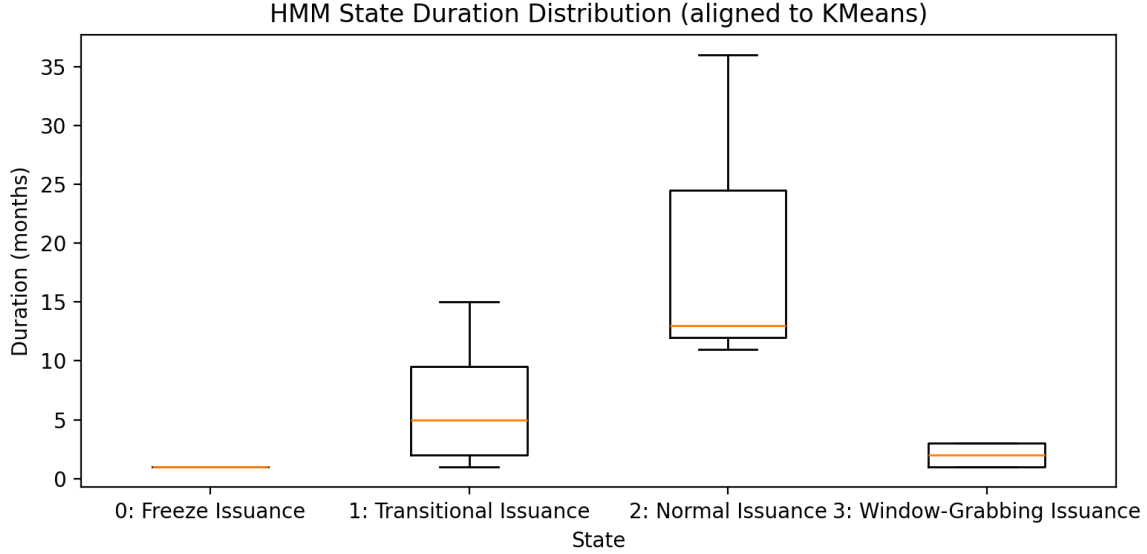


Figure 6: Distribution of contiguous episode durations by HMM regime.

Figure@fig-hmm-durations illustrates the distribution of contiguous episode lengths (in months) for each issuance regime.

Appendix E. Additional Macro–Regime Alignments

Issuance regime classification in this study is driven exclusively by issuance structure features. Macro-financial variables are introduced only ex post to provide economic context and to illustrate how different issuance regimes align with commonly used indicators of financial conditions. This appendix presents additional macro alignments and representative episodes for descriptive purposes.

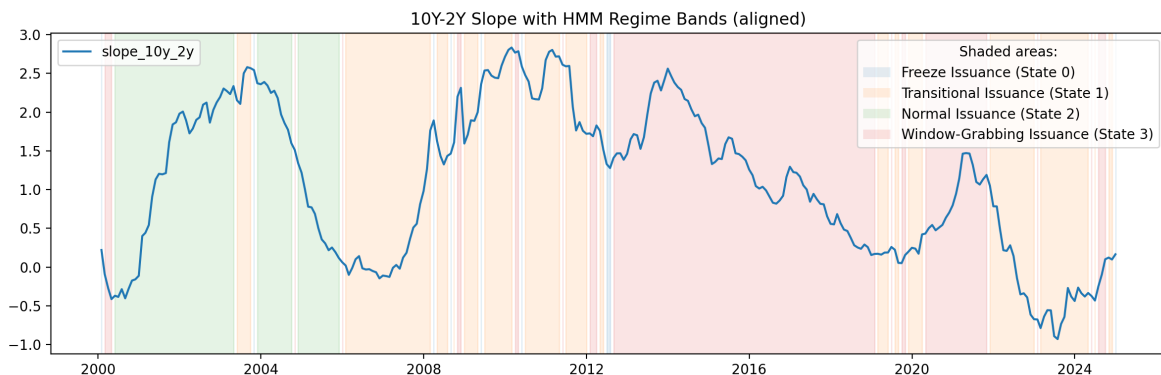


Figure 7: 10Y–2Y Treasury slope with HMM regime bands (aligned).

E.1 Yield-curve slope

Figure@fig-slope overlays the 10-year minus 2-year Treasury yield slope on the HMM regime bands. Periods with a steep, positive yield curve are frequently associated with the Normal Issuance regime. As the yield curve flattens or approaches inversion, issuance behavior more often transitions into Transitional or Window-Grabbing regimes, reflecting adjustment in issuance timing and coordination. Freeze Issuance episodes, which are rare, tend to occur when the yield curve is already flat or inverted and broader credit conditions are impaired.

E.2 High-yield credit risk

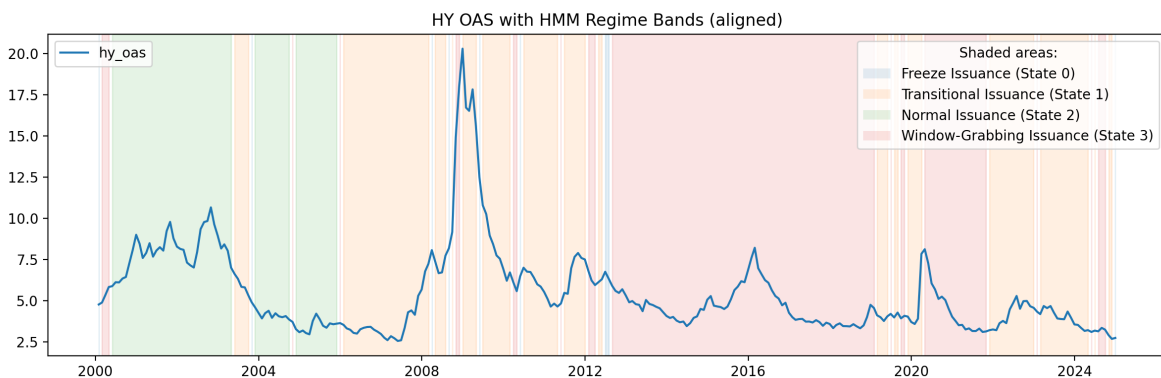


Figure 8: High-yield option-adjusted spreads (HY OAS) with HMM regime bands (aligned).

High-yield option-adjusted spreads (HY OAS) provide a summary measure of credit market stress. Figure@fig-hy-oas overlays HY OAS with the issuance regime bands. Elevated and persistent HY OAS levels are commonly observed during Freeze and Transitional regimes,

while Normal Issuance is typically associated with moderate or declining spreads. Window-Grabbing episodes tend to coincide with periods of rapid change in HY OAS, including sharp widenings or abrupt reversals, consistent with issuers attempting to access markets during short-lived windows of opportunity.

E.3 Other macro variables

Other macro-financial variables—such as equity market volatility (VIX), the federal funds rate, alternative credit spread measures (e.g., Baa–Aaa), and industrial production growth—can be examined using the same regime-overlay approach. Across these indicators, Freeze and Transitional Issuance regimes are generally observed in environments of elevated market stress or heightened policy uncertainty, while Normal Issuance prevails during more benign macroeconomic conditions. Window-Grabbing regimes often appear near turning points, when macro indicators are adjusting rapidly rather than remaining at extreme levels. These patterns are descriptive and are not used for regime identification or inference.

E.4 Representative months and episodes

To illustrate typical manifestations of each regime, representative months are selected based on high posterior state probabilities from the HMM. Examples of Freeze Issuance include September 2008 (following the Lehman Brothers collapse), March 2020 (during the COVID-19 shock), and brief episodes during the 2023 banking-sector stress. Transitional Issuance episodes cluster around periods such as early 2002, mid-2009, and mid-2023, which correspond to adjustment phases following major disruptions. Normal Issuance dominates extended tranquil intervals, including much of 2004–2007 and 2013–2019. Window-Grabbing episodes appear in short bursts around dates such as January 2001, May 2003, April 2009, and parts of 2020–2021, characterized by extreme temporal compression and concentrated issuer participation.

These illustrative examples underscore that similar macro-financial environments can support markedly different issuance behaviors, depending on how firms coordinate market access over time.

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